ORIGINAL ARTICLE

Workload and occupational stress as determinants of workplace violence among hospital nurses: A cross-sectional correlational study

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ABSTRACT

Background and aim of the work: Workplace violence is a serious issue in healthcare services and can negatively impact safety, psychological well-being, and the quality of nurses' work. High workload and increased occupational stress can heighten nurses' vulnerability to an increased risk of violence in hospital environments. This study aims to analyze the relationship between workload and occupational stress concerning incidents of workplace violence among nurses.
Research design and methods: This cross-sectional correlational study was carried out at a public Type B hospital in Banda Aceh. Structured questionnaires were used to assess workload, occupational stress, and workplace violence among 207 staff nurses selected through probability sampling. These instruments, adapted from a previous study, were administered with informed consent over two weeks. The relationships among workload, occupational stress, and workplace violence were analyzed using univariate statistics and Spearman's rank correlation.
Results: The correlation analyses revealed that workload is significantly positively associated with workplace violence ($r = 0.466$, $p < 0.001$), indicating a moderate link where increased perceived workload corresponds with higher violence. Additionally, occupational stress showed a significant and stronger correlation with workplace violence ($r = 0.542$, $p < 0.001$).
Conclusions: High workload and occupational stress frequently raise the risk of workplace violence from patients, family members, or colleagues. This situation not only affects nurses' mental health but also compromises the quality of care and endangers the development of a patient safety culture in hospitals. These results

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underscore the importance of implementing effective workload management and stress reduction strategies to improve nurses' safety and well-being. (www.actabiomedica.it)

Key words: workload, occupational stress, workplace violence, nurses, hospitals

Introduction

Safe and quality healthcare services are fundamental elements in the global health system. However, to date, developing countries still face significant challenges in ensuring patient safety (1). The WHO, OECD, and the World Bank noted that one in ten patients experiences injuries while receiving care in hospitals, even in developed countries. In developing countries, there are approximately 134 million adverse events resulting in 2.6 million deaths each year due to unsafe services (2).

In Indonesia, various policies have been implemented to strengthen patient safety culture (3), including the National Hospital Accreditation Standards and regulations issued by the Indonesian Ministry of Health (4). Nurses, especially those working in direct service units, play a strategic role in the success of implementing this culture. However, these efforts are often hindered by high workload issues and prolonged work-related stress.

Excessive workload causes physical and psychological fatigue among nurses, reduces productivity, and affects the quality of (5). Meanwhile, poorly managed occupational stress has been shown to contribute to burnout and ineffective patient care (6,7). Occupational stress can be caused by many factors, including organizational pressure, interpersonal conflicts, and imbalanced work demands (8). Furthermore, high workload and work-related stress often lead to an increased risk of workplace violence from patients, families, and colleagues. Putra et al.'s study shows that verbal and physical violence against nurses in Aceh is relatively high, with many incidents going unreported due to a weak incident reporting system (9,10).

In various global studies, violence against nurses has become a grave issue. Its prevalence reaches 91% in the United States, 72% in China, and 58% in Australia (11). This condition not only disrupts nurses' psychological well-being but also directly impacts the quality of care and threatens the implementation of a patient safety culture in hospitals (12). Based on these conditions, it is essential to explore the relationship between workload and work-related stress with workplace violence to obtain a scientific overview that can be used to develop more responsive and adaptive health human resource management strategies in the hospital environment, particularly at RSUD Meuraxa Banda Aceh.

Methods

This study employed a quantitative approach with a descriptive correlational design using a cross-sectional framework to examine the relationships between nurse workload, occupational stress, and workplace violence at a single point in time (13). The study was conducted at RSUD Meuraxa, a Type B public hospital in Banda Aceh, Indonesia, characterized by high service demand and limited nursing resources.

The target population comprised all staff nurses working at RSUD Meuraxa. Eligible participants were permanent nurses (excluding contract or internship staff) with a minimum of one year of work experience and who consented to participate by completing the questionnaire. The sample size was calculated using the Slovin formula with a 5% margin of error, resulting in a required sample of 207 respondents from a total population of 428 nurses. Probability sampling was employed to ensure equal selection opportunities for all population members, and proportional stratified random sampling was used to allocate samples across clinical units in proportion to their size.

Data were collected using three structured questionnaires adapted from validated instruments. Nurse workload was measured using a 13-item questionnaire, which covered physical (4 items), psychological (5 items), and working time aspects (4 items), all rated on a four-point Likert scale. Occupational stress was assessed using a 35-item instrument, encompassing physical (13 items), psychological (17 items), and behavioral (5 items) dimensions (14). Workplace violence was measured using a questionnaire that assessed types of violence, frequency, perpetrators, and reporting mechanisms, employing both categorical scales and open-ended responses (9).

Data collection was conducted over two weeks. Questionnaires were distributed in hard-copy format at each nursing unit. Before data collection, the research team explained the study objectives and procedures, and written informed consent was obtained from all participants. Respondents were given up to two days to complete and return the questionnaires. Ethical approval and institutional permission were obtained from the hospital management and the relevant Health Research Ethics Committee.

Data processing involved systematic stages of editing, coding, data entry, and cleaning using SPSS version 25 to ensure data accuracy and integrity. Univariate analysis was performed to describe respondent characteristics and the distribution of workload, occupational stress, and workplace violence using frequencies, percentages, means, and standard deviations. Bivariate analysis was conducted to examine relationships between variables. Due to non-normal data distribution, Spearman's rank correlation test was applied to assess the strength and direction of associations between workload, occupational stress, and workplace violence.

Results

Characteristics of respondents

The attributes of the study participants are outlined in Table 1 below.

Table 1 presents the socio-demographic characteristics of all respondents. Among the 207 participants, 74.9% (155) are women. The most represented age group is 31-40 years old, making up 51.2% of respondents. The most prevalent educational background among nurses is a DIII in Nursing, with 70.5% or 146 individuals. Most participants have 1-5 years of work experience, accounting for 61.4% of respondents, or 127 individuals. Over half of the respondents (52.7% or 119) report a monthly income above the district or city minimum wage. They are employed under PPPK status, which constitutes 57%, and most are married (78.7% or 163 people). Additionally, 47.3% or 96 respondents work in inpatient wards. A significant number (169, or 81.6%) have experienced WPV procedures at the hospital, while 76.8% (159) believe WPV handling training is unnecessary.

Nurse workload level

Table 2 shows the level of nurses' workload, where 59 people (28.5%) are classified as having severe workloads. Meanwhile, 95 respondents (45.9%) had moderate workloads, and the remaining 53 people (25.6%) were in the Mild workload category.

Nurse occupational stress level

Table 3 shows that the majority of respondents (160 people, 77.3%) fell into the category of Mild occupational stress. Respondents with Moderate occupational stress numbered 33 people (15.9%), while only 14 people (6.8%) experienced severe occupational stress.

Types of violence and perpetrators of violence in the workplace

The findings indicate that workplace violence experienced by nurses was predominantly non-physical in nature. Emotional abuse emerged as the most frequently reported form, with more than half of respondents experiencing low (31.9%) to intermediate (21.3%) levels, followed by threats (low 23.7% and intermediate 11.1%). In contrast, physical assault and verbal sexual harassment were relatively infrequent, while sexual abuse was almost absent. The mean scores across violence types further reinforce that emotional abuse and threats constitute the most salient forms of workplace violence in this setting (Table 4).

Table 1. Characteristics of respondents (n= 207).

No	Characteristics	Frequency (F)	Percentage (%)
1	**Gender**		
	Male	52	25.1
	Female	155	74.9
2	**Age**		
	20 – 30 years	74	35.7
	31 – 40 years	106	51.2
	41 – 50 years	26	12.6
	>50 years	1	0.5
3	**Educational Level**		
	Diploma	155	73.9
	Bachelor	47	22.7
	Magister	7	3.4
4	**Employment**		
	Civil Employee	20	9.7
	PPPK	118	57
	Contract	69	33.3
5	**Income**		
	Under is the City Minimum Wage	98	47.3
	Above the City Minimum Wage	109	52.7
6	**Marital Status**		
	Unmarried	41	19.8
	Marry	163	78.7
	Widow/Widowed	3	1.5
7	**Working Period**		
	1 – 5 years	127	61.4
	6 – 10 years	42	20.3
	16 – 15 years	21	10.1
	16 – 20 years	16	7.7
	>20 years	1	0.5
8	**Working Unit**		
	Emergency Department	20	9.7
	Intensive Care	41	19.8
	Polyclinic	12	5.8
	Inpatient Ward	134	64.7
9	**Mechanism/procedure for reporting WPV actions**		
	There is	169	81.6
	No	38	18.4
10	**Information or training on handling WPV**		
	There is	48	23.2
	No	159	76.8
11	**Concern about WPV Actions**	(Min-Max=1-5) M=3.52±1.31	
12	**Satisfaction with WPV handling procedures**	(Min-Max=1-5) M=2.99±1.12	

Regarding perpetrators (Figure 1), violence was most commonly attributed to patients' relatives or family members (60.5%) and followed by patients (22%). This highlights the central role of patient–family interactions as a significant source of risk. Incidents involving coworkers, other healthcare staff, or supervisors were reported less frequently. Overall, most nurses reported experiencing workplace violence at a low level (Table 5). However, a notable proportion reported moderate exposure, underscoring the persistence of workplace violence as an occupational concern requiring systematic prevention and reporting mechanisms.

Table 2. Nurse workload (n=207).

Workload	F	%
Severe	59	28.5
Moderate	95	45.9
Mild	53	25.6
Total	**207**	**100**

Table 3. Nurse occupational stress level (n=207).

Occupational stress	F	%
Severe	14	6.8
Moderate	33	15.9
Mild	160	6.8
Total	**207**	**100**

The relationship between nurse workload and incidents of workplace violence

Spearman's rank correlation analysis (Table 6) revealed a statistically significant positive association between nurse workload and incidents of workplace violence ($r = 0.466$, $p < 0.001$). This finding indicates a moderate correlation, suggesting that higher perceived workload levels are associated with an increased frequency of workplace violence incidents among nurses. The result implies that excessive workload may create conditions that elevate exposure to violent events in clinical settings.

Table 4. Types of violence (n=207).

No.	Types of violence F %		
1	Physical Assault (PA): (Min-Max= 0-5; M=0.23±0.63)	F	%
	Never Low Intermediate High	176 20 5 6	85 9.7 2.4 2.9
2	Emotional Abuse (EA): (Min-Max= 0-5; M=1±0.97)		
	Never Low Intermediate High	79 66 44 18	38.2 31.9 21.3 8.7
3	Threat (T): (Min-Max= 0-5; M=0.55±0.804)		
	Never Low Intermediate High	129 49 23 6	62.3 23.7 11.1 2.9
4	Verbal Sexual Harassment (VSH): (Min-Max= 0-5; M=0.18±0.49)		
	Never Low Intermediate	180 17 10	87 8.2 4.8
5	Sexual Abuse (SA): (Min-Max= 0-5; M=0.00±0.7)		
	Never Low	206 1	99.5 0.5

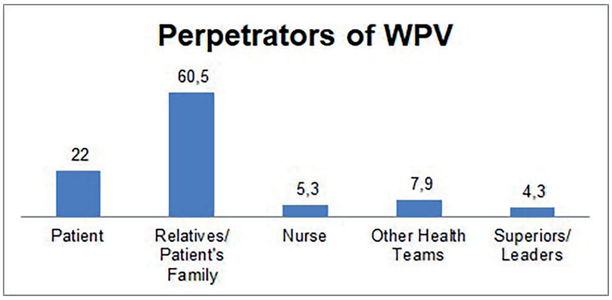

Figure 1. Perpetrators of workplace violence.

Table 5. Incidents of workplace violence among nurses (n=207).

Incident Report	F	%
Min-Max= 0-10; M=1.97±2.19		
Never	64	30.9
Low	125	60.4
Intermediate	18	8.7

Table 6. Relationship between nurse workload and incidents of workplace violence (n= 207).

Variable	R	p
Workload – Incident Workplace Violence	0.466	<0.001

The relationship between nurses' occupational stress and incidents of workplace violence

The results of Spearman's rank correlation analysis (Table 7) demonstrated a statistically significant and strong positive relationship between nurses' occupational stress and incidents of workplace violence ($r = 0.542$, $p < 0.001$). This finding indicates that higher levels of work-related stress are closely linked to greater exposure to workplace violence. The strength of this association underscores occupational stress as a critical factor contributing to workplace violence and highlights the importance of stress management and organizational support in violence prevention strategies.

Table 7. Relationship between nurses' occupational stress and incidents of workplace violence (n= 207).

Variable	R	p
Workload – Incident Workplace Violence	0.542	<0.001

Discussion

This study shows that most nurses experience a moderate workload. This result indicates that they face considerable physical and mental demands, though not at extreme levels. In nursing, workload encompasses the combined physical, mental, and emotional efforts involved in clinical and administrative tasks, as well as interactions with patients and families (14,15). The workload is dynamic and varies depending on the number of patients, their health conditions, and the complexity of nursing procedures at any moment.

Nurses' workload is affected by task volume, patient characteristics, nursing procedures, and the work environment. Repetitive activities requiring muscular effort and emotional involvement contribute to fatigue (16). Changes in activities during a shift, driven by fluctuations in patient numbers and conditions, cause variations in physical and mental workload, impacting fatigue levels (16). High workloads, such as imbalanced patient-nurse ratios and complex tasks, demand significant physical and psychological effort, increasing the risk of fatigue, occupational stress, and reduced professional performance (17).

Excessive workloads can cause numerous adverse effects, such as poor communication between nurses and patients, interprofessional miscommunication, increased nurse turnover, and decreased job satisfaction (18). Fatigue and stress from heavy workloads can impair nursing care quality, ultimately impacting service quality and patient safety (19). Consequently, regular workload assessments in each service area are essential, despite potential differences between units. Such evaluations are crucial for workforce planning, staff management, improving nursing care, and ensuring more equitable and effective resource distribution (16). Besides workload, this study indicates that most nurses experience mild occupational stress. Mild stress might act as an adaptive response, enhancing alertness and readiness to meet work demands. It is typically psychological and temporary, with individuals striving to complete tasks more quickly or efficiently, although it can gradually drain energy (20). Nonetheless, persistent occupational stress, even if mild, can develop into more serious stress if not adequately addressed (21).

Occupational stress is a response, either internal or external, that causes physical and psychological pressure beyond an individual's coping capacity, often manifesting as feelings of distress during work (22). In nurses, this stress can originate from various sources: anxiety about infection and death risks, demanding job requirements that keep them away from their families, exposure to traumatic events like patient deaths or critical illnesses, chronic heavy workloads, limited resources and staffing, as well as fatigue and burnout (7,23). Additional factors include job and environmental characteristics, lack of supervision, role conflict, long working hours, interpersonal issues, inadequate reward systems, poor organizational communication, and exposure to workplace violence (24). Moreover, occupational stress impacts nurses' social behaviors and professional performance, potentially leading to negative behaviors such as difficulties in collaboration, loss of interest, restlessness, and heightened startle responses, which affect workplace relationships. It has also been shown to influence the quality of care nurses provide (7,25). A study conducted at a hospital in Makassar found that nurses' occupational stress results from interactions between individual, organizational, and work environment factors (26).

Workplace violence in healthcare has been an ongoing issue and remains a significant global concern (27,28). The types and patterns of violence in healthcare settings are changing and becoming more complex (29). Violence against nurses can manifest as physical or verbal abuse, with perpetrators often being patients, their families, or even other healthcare workers. Previous research has documented incidents of physical assault and sexual harassment against nurses, which can have long-lasting physical and psychological effects (30). These effects include post-traumatic stress, reduced motivation at work, feelings of unfairness, and even a desire to leave the profession or workplace (31). Verbal abuse, the most common form of violence reported, includes rough treatment, inappropriate language, insults, and yelling (29). Nurses frequently face verbal abuse, not only from patients and their families but also from colleagues, supervisors, and other healthcare staff (32). In this study, most nurses reported experiencing low levels of workplace violence (60.4%), while 30.9% said they never experienced violence, and 8.7% reported moderate levels. While most reported only low levels, these findings still show that workplace violence remains a real issue for nurses.

According to the Occupational Safety and Health Administration, workplace violence encompasses any act or threat of physical violence, harassment, intimidation, or other threatening behavior occurring in the workplace (33). The healthcare industry exhibits a higher rate of violence compared to other sectors, with nurses being the most at risk of victimization (12). This finding is closely associated with nurses' roles as direct caregivers, involving frequent interaction with patients and their families, which increases their vulnerability to conflict and aggression (34). The correlation analysis in this study revealed a significant positive relationship between workload and workplace violence ($r = 0.466$; $p < 0.001$). These results corroborate previous studies indicating a high prevalence of violence, including exceptionally verbal assaults, among nurses under heavy workloads (9,10). Other research shows that verbal violence is the most common, followed by threats, physical violence, and sexual harassment (35). Heavy workloads, especially in resource-limited settings, heighten stress levels, hinder communication, and escalate conflicts, thereby raising the risk of violence (36,37). Excessive workloads can also impair nurses' communication with patients and families, leading to dissatisfaction and aggressive responses (17). Prior studies have found that nurses with high workloads face a greater risk of verbal violence than those with lighter workloads. Therefore, workload not only affects nurses' well-being but also directly influences the occurrence of violence in hospital settings.

This study also identified a significant positive link between occupational stress and workplace violence ($r = 0.542$; $p < 0.001$). This finding supports earlier research indicating that high occupational stress levels are associated with increased incidents of verbal and non-verbal violence against nurses. Occupational stress can lead to reduced performance, impaired concentration, fatigue, and strained interpersonal relationships, all of which may provoke conflicts and raise the risk of violence. Stress heightens emotional reactivity, making nurses more prone to irritability, loss of emotional control, and difficulty managing conflicts. Additionally, occupational stress can diminish assertive and therapeutic communication skills, increasing misunderstandings and verbal violence at work (38). Thus, addressing occupational stress is crucial in efforts to prevent workplace violence.

This study has several limitations. Its cross-sectional design prevents causal conclusions between the variables examined. Additionally, the study did not account for other factors like organizational culture, leadership style, shift patterns, and individual traits that could affect workplace violence. Conducted in a single healthcare facility, the study's findings have limited generalizability. Nonetheless, the results contribute valuable insights into the relationship between workload, job stress, and workplace violence among nurses. They can help guide the development of policies and interventions aimed at preventing violence in hospital settings.

Conclusion

This study shows that nurse workload and job stress are strongly linked to workplace violence among hospital nurses. Most nurses reported moderate workloads and mild stress; however, emotional abuse and threats were the most common types of violence, mainly coming from patients and their families. The results reveal a moderate positive relationship between workload and violence, and a stronger positive link between stress and violence. It suggests that higher work demands and psychological pressure increase nurses' risk of violence. These findings highlight that workplace violence is connected to organizational and psychosocial factors rather than being isolated incidents. Heavy workloads and unmanaged stress can impair communication, trigger emotional reactions, and escalate conflicts, harming nurses' well-being and patient safety. Hospital management should focus on appropriate staffing, regular workload reviews, stress reduction programs, and effective violence prevention and reporting systems to ensure a safe and supportive healthcare environment.

Ethics Approval: This research has passed the Ethical Feasibility Test from the Nursing Research Ethics Committee, Faculty of Nursing, Universitas Syiah Kuala, Research Code Number 113013240725, dated August 15, 2025.

Conflict of Interest: Each author declares that he or she has no commercial associations (e.g., consultancies, stock ownership, equity interest, patent/licensing arrangement, etc.) that might pose a conflict of interest in connection with the submitted article.

Authors' Contribution: MM systematically gathered data and developed the study's framework. MM and AP provided expert editorial insights, conducted thorough reviews of the concept, and oversaw the development of the article. AF played a significant role in drafting and advancing the idea. AM meticulously compiled the pertinent references and actively participated in the writing and editing of the paper. All authors collaboratively edited and revised the manuscript, ultimately reviewing and approving the final published version.

Declaration on the Use of AI: We began writing this paper in Indonesian. After the draft is finished, we translate using a tool, then synchronize each sentence and paragraph using Grammarly.

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References

1. Putra A, Kamil H, Yuswardi Y, Darmawati D. Frontline nursing leadership and safety culture in psychiatric care: Evidence from Aceh Mental Health Hospital. Acta Biomed Atenei Parm. 2025;96(3):16852. doi: 10.23750/abm.v96i3.16852

2. WHO, OECD, World Bank. Delivering Quality Health Services: A Global Imperative. OECD Publishing; 2018.
3. Kamil H, Yuswardi Y, Darmawati D, Putra A. Patient Safety Culture in Aceh Mental Hospital. J Liaquat Uni Med Heal Sci. 2025;Sp. Issue:38–43. doi: 10.22442/jlumhs.2025.01318
4. National Academies of Sciences, Medicine Division, Board on Global Health & C on I the Q of HCG. Crossing the global quality chasm: improving health care worldwide. 2018.
5. Piaster EH, Mardiyati SH, Kartika P, Hasanah UU, Al-Mubaroq MM. Effect of Workload on Fatigue in Nurses in The Hospital of Surakarta in 2023. J Mitra Kesehat. 2024;7(1):50–63. doi: 10.47522/jmk.v7i1.363
6. Devita M, Maryana, Nurvinanda R. Hubungan Burnout dan Stress Kerja Terhadap Kinerja Perawat Dirumah Sakit Umum Daerah Depati Hamzah Kota Pangkalpinang Tahun 2024. J Kesehat Tambusai. 2025;6(1):3759–66. Available from: https://journal.universitaspahlawan.ac.id/index.php/jkt/article/view/41250/27621
7. Putra A, Yuswardi Y, Nuraeni F. Factors influencing nurses' occupational stress while caring for COVID-19 patients: A retrospective study. Acta Biomed Atenei Parm. 2025;96(5):17009. doi: 10.23750/abm.v96i5.17009
8. Natasha CW, Putra A, Jannah N, Kamil H, Rachmah R. Beban Kerja Perawat Di Rumah Sakit Umum Daerah Kabupaten Aceh Jaya. J Ilm Mhs Fak Keperawatan. 2023;7(4).
9. Putra A, Kamil H, Adam M, Usman S. Prevalence of workplace violence in Aceh, Indonesia: A survey study on hospital nurses. Acta Biomed. 2024;95(1):e2024062. doi: 10.23750/abm.v95i1.15430
10. Putra A, Kamil H, Adam M, Usman S, Rauzatul Jannah S, Abdurrahman A. Workplace violence in healthcare: Insights into incidents, types, and reporting mechanisms for hospital nurses. Acta Biomed Atenei Parm. 2025;96(1):16356. doi: 10.23750/abm.v96i1.16356
11. Nelson S, Ayaz B, Baumann AL, Dozois G. A gender-based review of workplace violence amongst the global health workforce—A scoping review of the literature. Hammad N, editor. PLOS Glob Public Heal. 2024;4(7):e0003336. doi: 10.1371/journal.pgph.0003336
12. Putra A, Kamil H, Adam M, Usman S. Socio-Demographic and Workplace Violence among Nurses in Aceh, Indonesia: A Correlational Study. J Liaquat Univ Med Heal Sci. 2024;65–71. doi: 10.22442/jlumhs.2024.01127
13. Polit DF, Beck CT. Nursing research: Generating and assessing evidence for nursing practice. 10th ed. Philadelphia: Wolters Kluwer; 2017.
14. Nursalam. Manajemen Keperawatan: Aplikasi dalam Praktik Keperawatan Profesional. 4th ed. Jakarta: Salemba Medika; 2014. 117 p.
15. Retnosari DF, Dwiyanti E. Relation Between Workload and Nutritional Status With Fatigue Working of Outpatient Installation Nurses in RSI Jemursari [in Indonesian]. J Ilm Keperawatan (Scientific J Nursing). 2017 Mar 2;3 (1 SE-):8–17. Available from: https://journal.stikespemkabjombang.ac.id/index.php/jikep/article/view/2
16. Pillastrini P, Banchelli F, Guccione A, et al. Global Postural Reeducation in patients with chronic nonspecific neck pain: cross-over analysis of a randomized controlled trial. Med Lav. 2018;109(1):16–30. doi: 10.23749/mdl.v109i1.6677
17. Yildirim D, Aycan Z. Nurses' work demands and work–family conflict: A questionnaire survey. Int J Nurs Stud. 2008;45(9):1366–78. doi: 10.1016/j.ijnurstu.2007.10.010
18. Shah MK, Gandrakota N, Cimiotti JP, Ghose N, Moore M, Ali MK. Prevalence of and Factors Associated With Nurse Burnout in the US. JAMA Netw Open. 2021;4(2):e2036469–e2036469. doi: 10.1001/jamanetworkopen.2020.36469
19. Badri IA. Hubungan beban kerja dan lingkungan kerja dengan stres kerja perawat ruangan ICU dan IGD. Hum Care J. 2020;5(1):379. doi: 10.32883/hcj.v5i1.730
20. Henry JD, Crawford JR. The short-form version of the Depression Anxiety Stress Scales (DASS-21): Construct validity and normative data in a large non-clinical sample. Br J Clin Psychol. 2005;44(2):227–39. doi: 10.1348/014466505X29657
21. Salvagioni DAJ, Melanda FN, Mesas AE, González AD, Gabani FL, De Andrade SM. Physical, psychological and occupational consequences of job burnout: A systematic review of prospective studies. PLoS One. 2017;12(10):e0185781. doi: 10.1371/journal.pone.0185781
22. Afshari Saleh L, Niroumand S, Dehghani Z, Afshari Saleh T, Mousavi SM, Zakeri H. Relationship between workplace violence and work stress in the emergency department.. Vol. 12, Journal of Injury and Violence Research. 2020. p. 183–90. doi: 10.5249/jivr.vo112i2.1526
23. Adriaenssens J, De Gucht V, Maes S. Causes and consequences of occupational stress in emergency nurses, a longitudinal study. J Nurs Manag. 2015;23(3):346–58.
24. Alsharari AF, Abu-Snieneh HM, Abuadas FH, Elsabagh NE, Althobaity A, Alshammari FF, et al. Workplace violence towards emergency nurses: A cross-sectional multicenter study. Australas Emerg Care. 2022;25(1):48–54. doi: 10.1016/j.auec.2021.01.004
25. Swamy L, Mohr D, Blok A, et al. Impact of Workplace Climate on Burnout Among Critical Care Nurses in the Veterans Health Administration. Am J Crit Care. 2020;29(5):380–9. doi: doi.org/10.4037/ajcc2020831
26. Mulaindah D, Sahrul S. Gambaran stres kerja perawat IGD Rumah Sakit X yang ada di Makassar. J Psikol Ski (Sosial Klin Ind Organ. 2019;1(1):93–103. Available from: https://jurnal.uit.ac.id/JPS/article/view/170
27. Davey K, Ravishankar V, Mehta N, Ahluwalia T, Blanchard J, Smith J, et al. A qualitative study of workplace violence among healthcare providers in emergency departments in India. Int J Emerg Med. 2020;13(1):33. doi: 10.1186/s12245-020-00290-0
28. Liu J, Gan Y, Jiang H, et al. Prevalence of workplace violence against healthcare workers: a systematic review and meta-analysis. Occup Environ Med. 2019;76(12):927–37. doi: 10.1136/oemed-2019-105849
29. Putra A, Kamil H, Adam M. Understanding the factors related to workplace violence incidents among nurses in Aceh,

Indonesia. Med Hist. 2025;9(1–2):16423. doi: 10.69124/mh.v9i1-2.16423
30. García-Pérez MD, Rivera-Sequeiros A, Sánchez-Elías TM, Lima-Serrano M. Workplace violence on healthcare professionals and underreporting: Characterization and knowledge gaps for prevention. Enfermería Clínica (English Ed). 2021;31(6):390–5. doi: 10.1016/j.enfcle.2021.05.001
31. Kibunja BK, Musembi HM, Kimani RW, Gatimu SM. Prevalence and Effect of Workplace Violence against Emergency Nurses at a Tertiary Hospital in Kenya: A Cross-Sectional Study. Saf Health Work. 2021;12(2):249–54. doi: 10.1016/j.shaw.2021.01.005
32. Mahdarsari M, Maurissa A, Putra A. Exploring the Factors Related to Workplace Violence among Nurses in Two Districts of Aceh Province. J Liaquat Uni Med Heal Sci. 2025;Sp. Issue:57–63. doi: 10.22442/jlumhs.2025.01321
33. OSHA. Guidelines for preventing workplace violence for health care and social service workers. Prairie Rose. 2016;66(1 Suppl).
34. Al-Omari H. Physical and verbal workplace violence against nurses in Jordan. Int Nurs Rev. 2015;62(1):111–8. doi: 10.1111/inr.12170
35. Park M, Cho SH, Hong HJ. Prevalence and Perpetrators of Workplace Violence by Nursing Unit and the Relationship Between Violence and the Perceived Work Environment. J Nurs Scholarsh. 2015;47(1):87–95. doi: 10.1111/jnu.12112
36. Zhang SE, Liu W, Wang J, et al. Impact of workplace violence and compassionate behaviour in hospitals on stress, sleep quality and subjective health status among Chinese nurses: a cross-sectional survey. BMJ Open. 2018;8(10):e019373. doi: 10.1136/bmjopen-2017-019373
37. Edward KL, Ousey K, Warelow P, Lui S. Nursing and aggression in the workplace: a systematic review. Br J Nurs. 2014;23(12):653-4, 656-9. doi: 10.12968/bjon.2014.23.12.653
38. Schiavo R. Health communication: From theory to practice. John Wiley & Sons; 2013.

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